

THE LANCET

Neurology

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

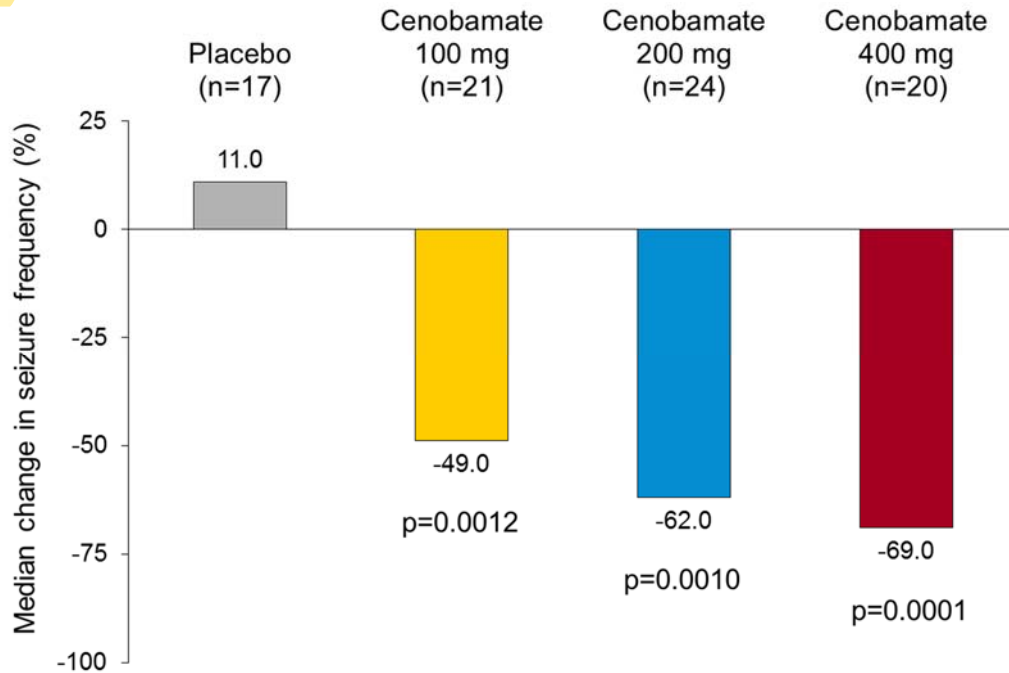
This online publication has been corrected. The corrected version first appeared at thelancet.com/neurology on Feb 18, 2020.

Supplement to: Krauss GL, Klein P, Brandt C, et al. Safety and efficacy of adjunctive cenobamate (YKP3089) in patients with uncontrolled focal seizures: a multicentre, double-blind, randomised, placebo-controlled, dose-response trial. *Lancet Neurol* 2019; published online Nov 13. [http://dx.doi.org/10.1016/S1474-4422\(19\)30399-0](http://dx.doi.org/10.1016/S1474-4422(19)30399-0).

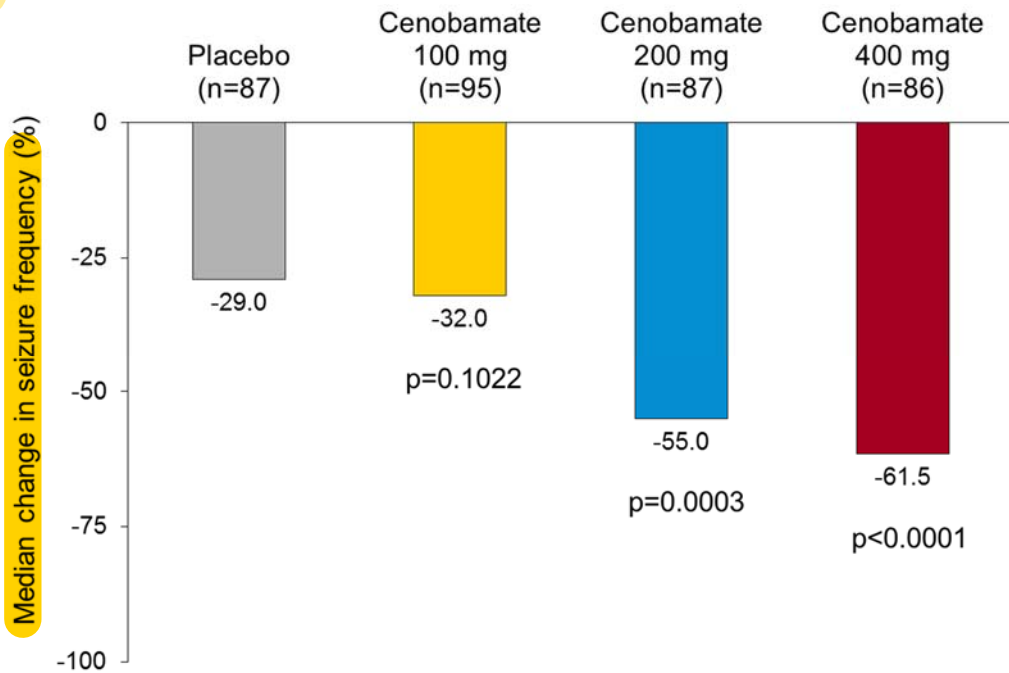
SUPPLEMENTARY APPENDIX

Supplementary Appendix figure 1: Median percent change in 28-day focal seizure frequency for focal seizure subtypes (MITT-M population, n=397). (A) Focal aware motor. (B) Focal impaired awareness. (C) Focal to bilateral tonic-clonic.

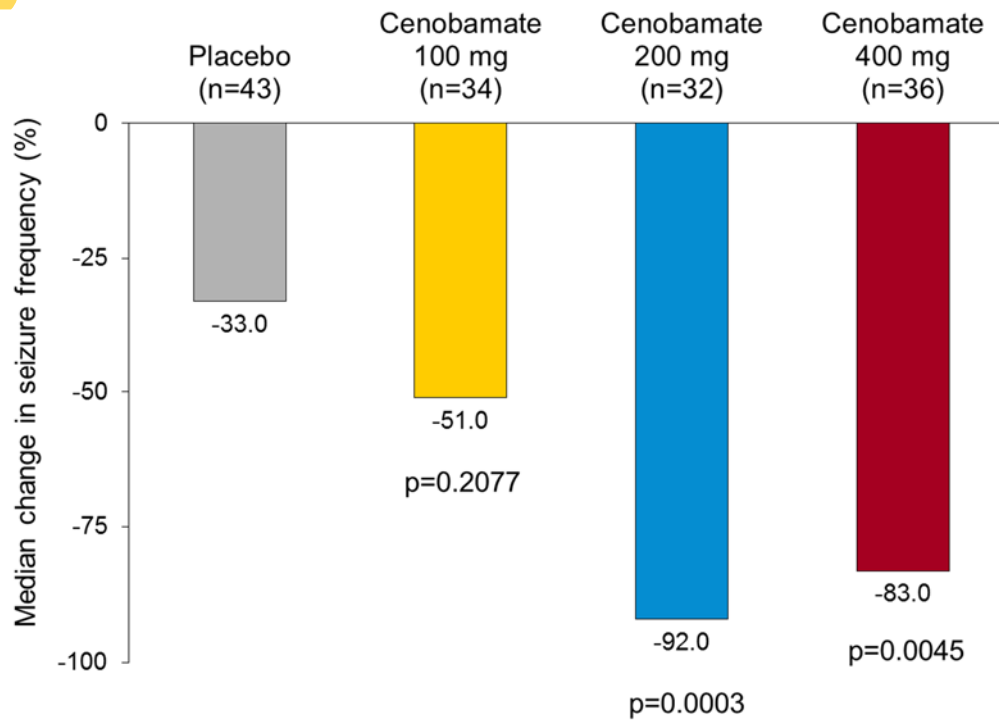
A.



B.



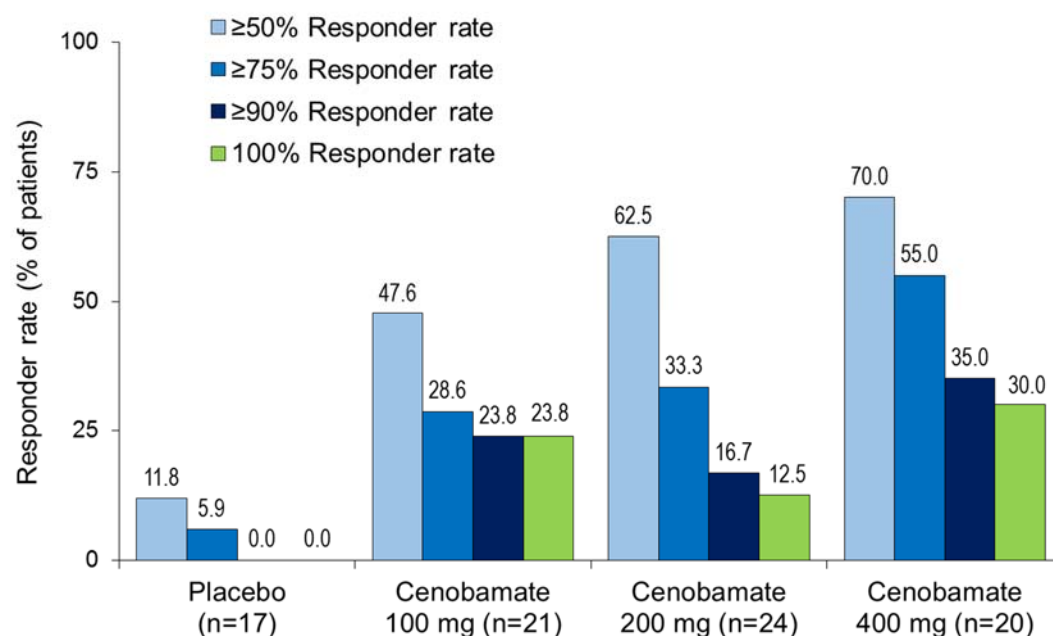
C.



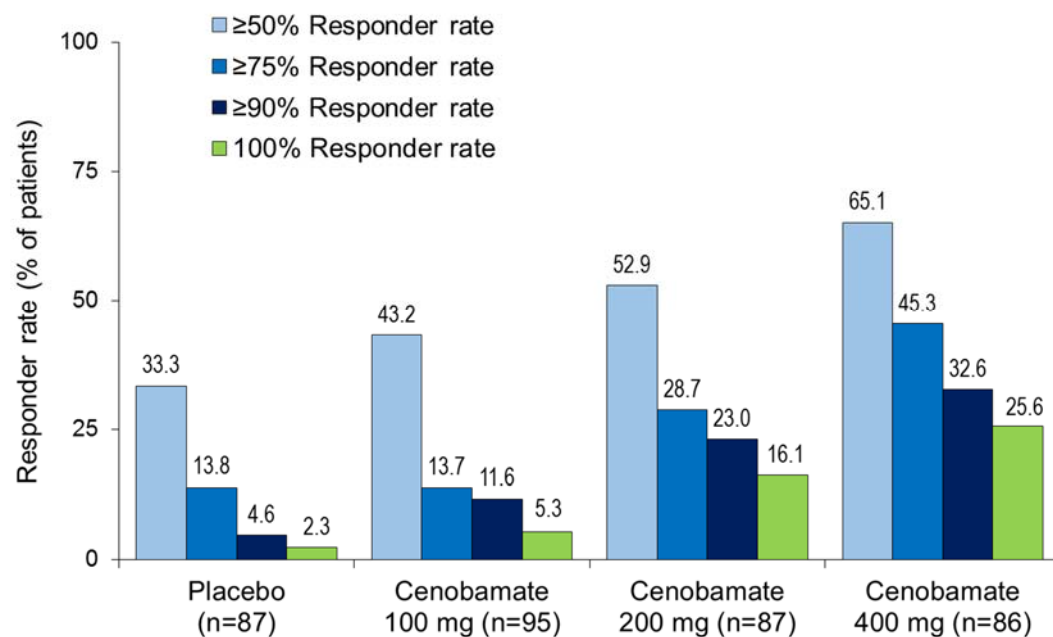
p-values (assessed post-hoc) are versus placebo.

Supplementary Appendix figure 2: Post-hoc responder rate for focal seizure subtypes during the maintenance phase (MITT-M population, n=397). (A) Focal aware motor. (B) Focal impaired awareness. (C) Focal to bilateral tonic-clonic.

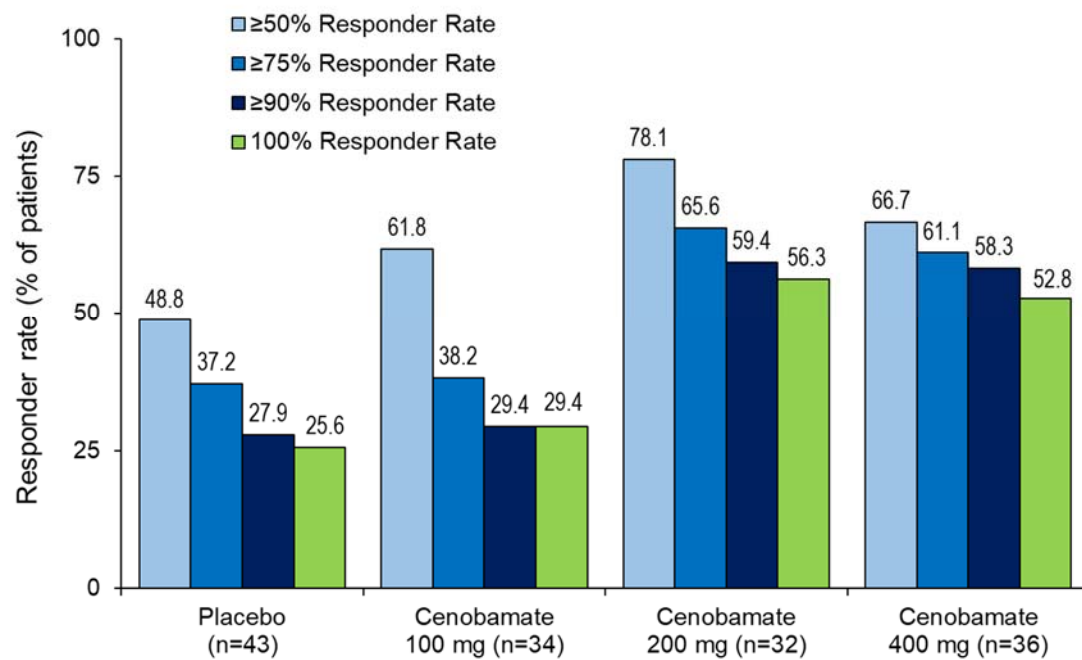
A.



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Supplemental Appendix table 1: Inclusion/exclusion criteria

Inclusion criteria
• Male or female, age 18 to 70 years, inclusive • Weight at least 40 kg • Written informed consent was signed by the patient or legal guardian prior to entering the study in accordance with the ICH GCP guidelines. If the written informed consent was provided by the legal guardian because the patient was unable to do so, a written or verbal assent from the patient was also to be obtained. • A diagnosis of focal epilepsy according to the International League Against Epilepsy's Classification of Epileptic Seizures. Diagnosis should have been established by clinical history and an EEG that was consistent with localization-related epilepsy; normal interictal EEGs were allowed provided that the subject met the other diagnosis criterion (ie, clinical history). • Uncontrolled focal seizures despite having been treated with at least one AED within approximately the last 2 years. • During the 8-week baseline period, subjects must have had at least 8 partial seizures including only focal aware seizures with motor component, focal unaware seizures, or focal to bilateral tonic-clonic seizures without a seizure-free interval of >25 days any time during the 8 weeks baseline. Subjects must have had at least three of these partial seizures during each of the two consecutive 4-week segments of the baseline period. • Currently on stable antiepileptic treatment regimen: - Subject must have been receiving stable doses of one to three AEDs for at least 4 weeks prior to screening (Visit 1) to be continued unchanged throughout the study. - Vagal nerve stimulator was not counted as an AED; however, the parameters were to be stable for at least 4 weeks prior to baseline. The vagal nerve stimulator must have been implanted at least 5 months prior to Visit 1. - Benzodiazepines taken at least once per week during the 1 month prior to Visit 1 for epilepsy, or for anxiety or sleep disorder, was counted as one AED and had to be continued unchanged throughout the study. Therefore, only a maximum of two additional approved AEDs were allowed. • CT or MRI scan performed within the past 10 years that ruled out a progressive cause of epilepsy. If a CT or MRI scan had not been performed within the past 10 years, one had to be performed prior to randomization. • Patient had to be available for telephone contact • Use of an acceptable form of birth control by female patients of childbearing potential

Exclusion criteria

- History of serious systemic disease, including hepatic insufficiency, renal insufficiency, a malignant neoplasm, any disorder in which prognosis for survival was less than 3 months, or any disorder that in the judgment of the investigator would have placed the subject at excessive risk by participation in a controlled trial.
- History of non-epileptic or psychogenic seizures
- Presence of only focal aware non-motor seizures or primary generalized epilepsies
- History of seizure clusters (episodes lasting less than 30 minutes in which multiple seizures occurred with such frequency that the initiation and completion of each individual seizure could not be distinguished) within 3 months prior to Visit 1.
- Presence or previous history of Lennox-Gastaut syndrome
- Scheduled epilepsy surgery within 8 months after Visit 1
- Pregnant or lactating
- Any clinically significant laboratory abnormality that in the opinion of the investigator would exclude the patient from the study
- Evidence of significant active hepatic disease. Liver transaminases (AST or ALT) above twice the upper limit of normal or total or direct bilirubin not within normal limits.
- Active CNS infection, demyelinating disease, degenerative neurologic disease, or any CNS disease deemed to be progressive during the course of the study that may have confounded the interpretation of the study results
- Any clinically significant psychiatric illness or psychological or behavioral problems that, in the opinion of the investigator, would have interfered with the subject's ability to participate in the study
- Presence of psychotic disorders and/or unstable recurrent affective disorders evident by use of antipsychotics; presence or recent history (within 6 months) of major depressive episode
- Used intermittent rescue benzodiazepines more than once per month (1 to 2 doses in a 24-hour period was considered 1 rescue) in the 1 month period prior to Visit 1
- History of alcoholism, drug abuse, or drug addiction within the past 2 years
- Current use of felbamate with less than 18 months of continuous exposure
- Current use of diazepam, phenytoin, phenobarbital, or metabolites of these drugs (within 1 month of Visit 1)
- Current or recent (within the past year) use of vigabatrin. Subjects with a prior history of treatment with vigabatrin must have had documentation showing no evidence of a vigabatrin-associated clinically significant abnormality in a visual perimetry test.
- History of status epilepticus within 3 months of Visit 1
- History of 1 serious drug-induced hypersensitivity reaction (including but not limited to Stevens Johnson syndrome, toxic epidermal necrolysis, DRESS) or any drug-related rash requiring hospitalization
- History of AED-associated rash that involved conjunctiva or mucosae or more than 1 maculopapular rash that required discontinuation
- Creatinine clearance less than 50 mL/min, as calculated by the Cockcroft-Gault equation

- Absolute neutrophil count less than 1500/ μ L
 - Clinical or ECG evidence of serious cardiac disease, including ischemic heart disease, uncontrolled heart failure, and major arrhythmias or relevant replicated changes in QT intervals (QT interval corrected for heart rate using Fridericia's formula [QTcF] <340 msec or >450 msec in males and >470 msec in females)
 - Platelet counts lower than 80,000/ μ L in subjects treated with valproic acid
 - Answered "yes" to Question 1 or 2 of the C-SSRS (baseline/screening version) ideation section in the past 6 months or "yes" to any of the suicidal behavior questions in the past 2 years
 - More than 1 lifetime suicide attempt
 - Participated in any other studies involving an investigational product or device within 30 days of screening (or longer, as required by local regulations)
 - Current use of any of the following medications: clopidogrel, fluvoxamine, amitriptyline, clomipramine, bupropion, methadone, ifosfamide, cyclophosphamide, efavirenz, or natural progesterone (within 1 month of Visit 1)
 - History of a positive antibody/antigen test for hepatitis B, hepatitis C, or human immunodeficiency virus
 - Presence of congenital short QT syndrome
 - History of any previous exposure to cenobamate
 - Any condition(s) that would make the subject, in the opinion of the investigator, unsuitable for the study (applied to sites in France only)
 - Persons under tutelage of guardianship-protected people in accordance with French regulations (applied to sites in France only)
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AED=antiepileptic drug; ALT=alanine aminotransferase; AST=aspartate aminotransferase; CNS, central nervous system; C-SSRS= Columbia-Suicide Severity Rating Scale; CT=computed tomography; DRESS= drug reaction with eosinophilia with systemic symptoms; ECG, electrocardiogram; EEG=electroencephalogram; ICH GCP= International Conference on Harmonisation Good Clinical Practice; MRI=magnetic resonance imaging.